

DHT Sensor integration with Thingspeak Cloud

Step 1: Connect Hardware

Wire the DHT11 sensor to your Raspberry Pi 4 GPIO header:

- **VCC** $\rightarrow$ **Pin 1** (3.3V) or **Pin 2** (5V)
- **GND** $\rightarrow$ **Pin 6** (Ground)
- **DATA** $\rightarrow$ **Pin 7** (GPIO 4)

(Note: If using a 4-pin raw DHT11 sensor without a breakout board, place a 10k Ω pull-up resistor between the VCC and DATA pins).

Step 2: Set Up ThingSpeak (Free Account)

1. Go to [ThingSpeak.com](https://thingspeak.com) and sign up for a free account.
2. Click **Channels** $\rightarrow$ **My Channels** $\rightarrow$ **New Channel**.
3. Configure the channel:

- **Name:** Raspberry Pi Weather
- **Field 1:** Temperature
- **Field 2:** Humidity

4. Click **Save Channel** at the bottom.
 5. Go to the **API Keys** tab and copy your **Write API Key**. You will paste this key into the Python script.
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Step 3: Install Required Packages on Raspberry Pi 4

Open the terminal on your Raspberry Pi 4 and set up a Python virtual environment to install the required libraries:

```
# Update system packages
sudo apt update && sudo apt upgrade -y

# Install required system dependencies
sudo apt install python3-pip python3-venv libgpiod2 -y

# Create and activate a Python virtual environment
python3 -m venv env
source env/bin/activate
```

```
# Install Adafruit DHT library and Requests library
pip install adafruit-circuitpython-dht requests
```

Step 4: Python Script to Send Data to ThingSpeak

Create a Python script named `thingspeak_dht11.py` :

```
nano thingspeak_dht11.py
```

Paste the following Python code into the file (make sure to replace `'YOUR_WRITE_API_KEY'` with the key from Step 2):

```
import time
import board
import adafruit_dht
import requests

# 1. Configuration
# Paste your ThingSpeak Write API Key here
THINGSPEAK_API_KEY = "YOUR_WRITE_API_KEY"
THINGSPEAK_URL = "https://api.thingspeak.com/update"

# 2. Sensor Setup (DHT11 connected to GPIO 4)
dht_device = adafruit_dht.DHT11(board.D4)

def send_to_thingspeak(temp, humidity):
    """Sends temperature and humidity readings to ThingSpeak API."""
    payload = {
        "api_key": THINGSPEAK_API_KEY,
        "field1": temp,
        "field2": humidity
    }

    try:
        response = requests.post(THINGSPEAK_URL, data=payload, timeout=10)
        if response.status_code == 200 and response.text != "0":
            print(f>Data sent successfully -> Temp: {temp}°C | Humidity: {humidity}%")
            return True
        else:
            print(f>ThingSpeak API Error. Status Code: {response.status_code}, Response: {resp
            return False
    except Exception as e:
        print(f>Connection error: {e}")
        return False
```

```

def main():
    print("Starting DHT11 to ThingSpeak Uploader (Press Ctrl+C to stop)...")

    while True:
        try:
            # Read sensor values
            temperature = dht_device.temperature
            humidity = dht_device.humidity

            if temperature is not None and humidity is not None:
                # Transmit data to ThingSpeak
                send_to_thingspeak(temperature, humidity)
            else:
                print("Failed to retrieve data from sensor. Retrying...")

        except RuntimeError as error:
            # DHT11 sensors frequently throw minor timing errors; catch and continue
            print(f"Sensor read error: {error.args[0]}")
        except Exception as error:
            dht_device.exit()
            raise error

        # ThingSpeak free tier requires at least 15 seconds between updates
        time.sleep(20)

if __name__ == "__main__":
    try:
        main()
    except KeyboardInterrupt:
        print("\nProgram stopped by user.")
    finally:
        dht_device.exit()

```

Save and exit `nano` by pressing `Ctrl+O` , `Enter` , then `Ctrl+X` .

Step 5: Run & Verify

Run the script from inside your virtual environment:

```
python3 thingspeak_dht11.py
```

- Go to your **ThingSpeak Channel** $\rightarrow$ **Private View** tab.
- You will see two live charts updating every 20 seconds with real-time temperature and humidity readings from your Raspberry Pi 4.